



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2020 Year 12 Course Assessment Task 3

Tuesday, 30 June 2020

General instructions

- Working time – 60 minutes.
(plus 5 minutes reading time)
- Commence each new question on a new Writing Booklet. Write on both sides of the paper.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, arrange the booklets used in the correct order after this paper and hand to examination supervisors.
- A NESA Reference Sheet is provided.

Class (please ✓)

- ☐ 12MXX.1 – Ms Ham
- ☐ 12MXX.2 – Mr Lam

NESA STUDENT #: # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	Total	%
MARKS	$\overline{13}$	$\overline{17}$	$\overline{10}$	$\overline{40}$	

Question 1 (13 Marks)**Marks**

- (a) Prove that 3

A three digit number is divisible by 9 if and only if the sum of its digits is divisible by 9.

Hint: Let the digits be a , b and c .

- (b) Prove the following statement by contrapositive: 2

For $n \in \mathbb{Z}$, if $n^2 - 6n + 5$ is even, then n is odd.

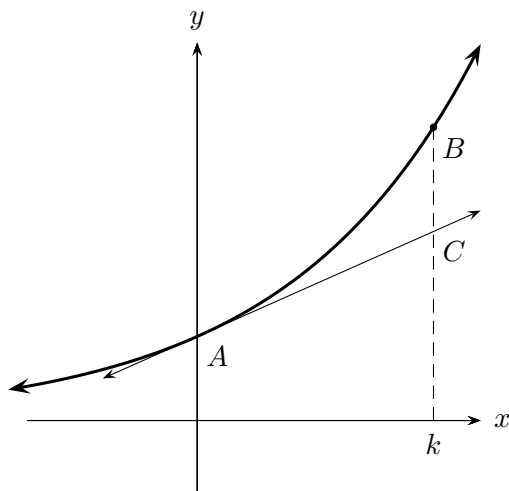
- (c) If a , b , c and d are all positive real numbers,
i. Show that 2

$$a^2 + b^2 + c^2 + d^2 \geq \frac{2}{3} (ab + ac + ad + bc + bd + cd)$$

- ii. Hence or otherwise, show that 2

$$\left(\frac{a + b + c + d}{4} \right)^2 \geq \frac{1}{6} (ab + ac + ad + bc + bd + cd)$$

- (d) The curve $y = e^x$ cuts the y axis at A . B is a second point on the curve such that $x = k$, where $k > 0$. The tangent to the curve $y = e^x$ at A cuts the vertical line $x = k$ at the point C .



- i. By considering areas, show that 3

$$\frac{1}{2}k(k+2) < e^k - 1 < \frac{1}{2}k(1 + e^k)$$

- ii. Hence deduce that $2.5 < e < 3$. 1

Question 2 (17 Marks)**Marks**

- (a) i. Find the partial fraction decomposition of $\frac{1}{x^2 - 1}$. **2**

- ii. Hence or otherwise, evaluate $\int \frac{x^2 + 1}{x^2 - 1} dx$. **2**

- (b) Use the substitution $x = \tan^2 \theta$ to evaluate **4**

$$\int_0^3 \frac{\sqrt{x}}{(1+x)^{\frac{5}{2}}} dx$$

Expressing your answer in simplest exact form.

- (c) Use the substitution $t = \tan \frac{x}{2}$ to evaluate **4**

$$\int_0^{\frac{\pi}{2}} \frac{1}{2 - \cos x + 2 \sin x} dx$$

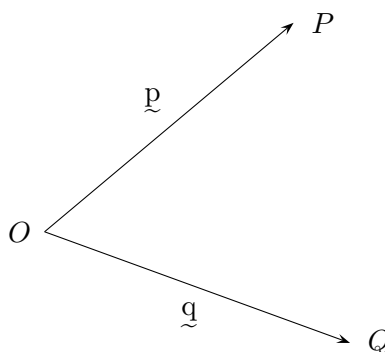
- (d) i. Given $I_n = \int_0^1 x^n 2^x dx$ where $n \in \mathbb{Z}^+$, show that **2**

$$I_n = \frac{2}{\ln 2} - \frac{n}{\ln 2} I_{n-1}$$

- ii. Hence evaluate $\int_0^1 x^3 2^x dx$. **3**

Question 3 (10 Marks)**Marks**

- (a) In the following diagram, let $\overrightarrow{OP} = \underline{p}$ and $\overrightarrow{OQ} = \underline{q}$.



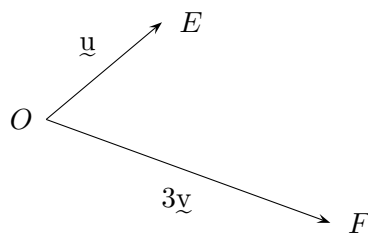
- i. Reproduce the diagram on to your writing booklet. In addition, draw the vector **1**

$$\overrightarrow{OR} = \underline{p} + \underline{q}$$

- ii. State the conditions on $|\underline{p}|$ and $|\underline{q}|$ such that $\overrightarrow{OR}$ bisects $\angle POQ$, giving brief reason(s). **2**

Examination continues overleaf...

- (b) The following figure shows $\overrightarrow{OE} = \underline{u} = \begin{pmatrix} 2 \\ 5 \\ -7 \end{pmatrix}$ and $\overrightarrow{OF} = 3\underline{v} = \begin{pmatrix} 15 \\ 21 \\ 6 \end{pmatrix}$.



- i. Find a vector $\overrightarrow{OG}$ in column vector notation that bisects $\angle EOF$. **2**
- ii. State the name of the shape formed by $OEGF$, and a brief reason for why $OEGF$ forms this shape. **2**
- iii. Show that the area A of, $OEGF$ is **3**

$$A = 2\sqrt{5123}$$

Hint: Consider $\text{proj}_{\underline{v}} \underline{u}$.

End of paper.

Suggested Band E4 Responses

Question 1

(a) (3 marks)

- ✓ [1] to define the sufficient and necessary conditions.
- ✓ [1] for proving the forward statement.
- ✓ [1] for proving the converse statement.
- Let the number n be

$$100a + 10b + c$$

such that a, b, c are single digit numbers from 0 to 9 with $a \neq 0$.

– Let p be the statement

The number $100a + 10b + c$ is divisible by 9

– Let q be the statement

The sum of its digits $a + b + c$ is divisible by 9

- **Forward statement:** $p \rightarrow q$
If

$$n = 100a + 10b + c$$

is divisible by 9, then

$$100a + 10b + c = 9A$$

where $A \in \mathbb{Z}^+$, then

$$\begin{aligned} & 100a + 10b + c - (99a + 9b) \\ &= 9A - 9(11a + b) \\ &= 9(A - 11a + b) \\ &= 9B \end{aligned}$$

where $B \in \mathbb{Z}$ is also divisible by 9. Therefore,

$$100a + 10b + c - (99a + 9b) = a + b + c$$

is divisible by 9, i.e. the sum of its digits $a + b + c$ is divisible by 9.

- **Converse statement:** $q \rightarrow p$
If $a + b + c$ is divisible by 9, then

$$\begin{aligned} a + b + c &= 9A \quad A \in \mathbb{Z}^+ \\ a + b + c + (99a + 9b) & \\ &= 9A + 99a + 9b \\ &= 9(A + 11a + b) \end{aligned}$$

is also divisible by 9, as $99a + 9b = 9(11a + b)$ is divisible by 9. Therefore,

$$a + b + c + 99a + 9b = 100a + 10b + c$$

is divisible by 9, i.e. n is divisible by 9.

Hence $p \leftrightarrow q$, and a number is divisible by 9 iff the sum of its digits is divisible by 9.

(b) (2 marks)

- Let p be the statement
 $n^2 - 6n + 5$ is even
- Let q be the statement
 n is an odd number

Required to prove:

$$\neg q \rightarrow \neg p$$

i.e. if n is not an odd number, then $n^2 - 6n + 5$ is not even, or

If n is an even number, then $n^2 - 6n + 5$ is odd.

Proof Let $n = 2k$ where $k \in \mathbb{Z}$.

$$\begin{aligned} & n^2 - 6n + 5 \\ &= (2k)^2 - 6(2k) + 5 \\ &= 4k^2 - 12k + 5 \\ &= 2(2k^2 - 6k + 2) + 1 \\ &= 2B + 1 \end{aligned}$$

where $B \in \mathbb{Z}$. $\therefore \neg q \rightarrow \neg p$. The contrapositive of the statement is true, hence

$$p \rightarrow q$$

is a true statement.

i. (2 marks)

- ✓ [1] Uses $(a - b)^2 \geq 0$ and its subsequent derivation for $a^2 + b^2 \geq 2ab$.
- ✓ [1] Uses the fact, enumerates through pairs and deduces the result.

$$\text{As } (a - b)^2 = a^2 - 2ab + b^2 \geq 0,$$

$$\therefore a^2 + b^2 \geq 2ab \quad (1.1)$$

Similarly,

$$a^2 + c^2 \geq 2ac \quad (1.2)$$

$$a^2 + d^2 \geq 2ad \quad (1.3)$$

$$b^2 + c^2 \geq 2bc \quad (1.4)$$

$$b^2 + d^2 \geq 2bd \quad (1.5)$$

$$c^2 + d^2 \geq 2cd \quad (1.6)$$

Add (1.1) to (1.6) and factorising,

$$\begin{aligned} & 3(a^2 + b^2 + c^2 + d^2) \\ & \geq 2(ab + ac + ad + bc + bd + cd) \end{aligned}$$

$$\begin{aligned} & \therefore a^2 + b^2 + c^2 + d^2 \\ & \geq \frac{2}{3}(ab + ac + ad + bc + bd + cd) \end{aligned}$$

ii. (2 marks)

- ✓ [1] Correctly expands $(a + b + c + d)^2$.
- ✓ [1] Uses the expansion and deduces the result.

$$\begin{aligned} & (a + b + c + d)^2 \\ & = a^2 + b^2 + c^2 + d^2 \\ & + 2(ab + ac + ad + bc + bd + cd) \end{aligned} \quad (1.7)$$

From part (i) and (1.7),

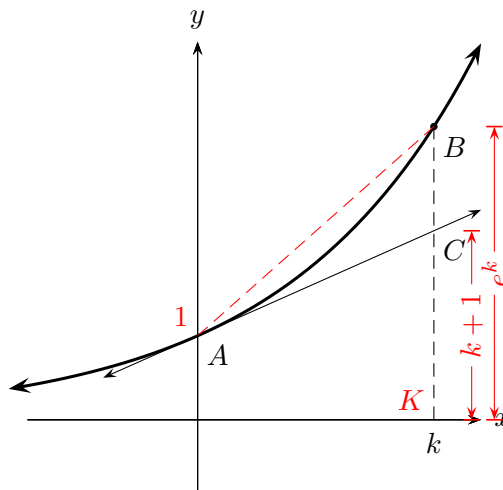
$$\begin{aligned} & (a + b + c + d)^2 \\ & \geq \frac{2}{3}(ab + ac + ad + bc + bd + cd) \\ & + 2(ab + ac + ad + bc + bd + cd) \\ & = \frac{8}{3}(ab + ac + ad + bc + bd + cd) \end{aligned}$$

Dividing $(a + b + c + d)^2$ by 16,

$$\begin{aligned} & \left(\frac{a + b + c + d}{4} \right)^2 \\ & \geq \frac{1}{16} \times \frac{8}{3}(ab + ac + ad + bc + bd + cd) \\ & = \frac{1}{6}(ab + ac + ad + bc + bd + cd) \end{aligned}$$

(d) i. (3 marks)

- ✓ [1] Finds the lower bound for $e^k - 1$ by comparing the definite integral with the underestimate.
- ✓ [1] Find the upper bound for $e^k - 1$ by comparing the definite integral with the overestimate.
- ✓ [1] For comparing the exact area with the lower/upper bounds, and concluding the result.



- Equation of tangent at A:

$$\begin{aligned} \frac{dy}{dx} &= e^x \Big|_{x=0} = 1 \\ \therefore y &= x + 1 \end{aligned}$$

At $x = k$,

$$\begin{aligned} y &= x + 1 \Big|_{x=k} = k + 1 \\ \therefore C &= (k, k + 1) \end{aligned}$$

- Underestimate of area: trapezium AOKC, where O is the origin:

$$\begin{aligned} A_{AOKC} &= \frac{1}{2}h(a + b) \\ &= \frac{1}{2}(k)(1 + k + 1) \\ &= \frac{1}{2}k(k + 2) \end{aligned}$$

- Overestimate of area: trapezium AOKB, where O is the origin:

$$\begin{aligned} A_{AOKB} &= \frac{1}{2}h(a + b) \\ &= \frac{1}{2}(k)(1 + e^k) \end{aligned}$$

- Exact area:

$$\begin{aligned} A &= \int_0^k e^x dx \\ &= [e^x]_0^k \\ &= e^k - e^0 \\ &= e^k - 1 \end{aligned}$$

The exact area is bounded by the underestimate and overestimate:

$$\begin{aligned} A_{AOKC} &< \int_0^k e^x dx < A_{AOKB} \\ \frac{1}{2}k(k+2) &< e^k - 1 < \frac{1}{2}k(1+e^k) \end{aligned}$$

- ii. (1 mark) When $k = 1$,

$$\begin{aligned} \frac{1}{2}(3) + 1 &< e^1 < \frac{1}{2}(1+e) + 1 \\ \therefore \frac{5}{2} &< e < \underbrace{\frac{1}{2}(3+e)}_{\text{examine this}} \\ \frac{2e}{-e} &< 3 + \frac{e}{-e} \\ \therefore e &< 3 \\ \therefore 2.5 &< e < 3 \end{aligned}$$

Question 2

- (a) i. (2 marks)

✓ [1] for each of A and B .

$$\begin{aligned} \frac{1}{(x-1)(x+1)} &\equiv \frac{A}{x-1} + \frac{B}{x+1} \\ \frac{1}{\times(x-1)(x+1)} &\times(x-1)(x+1) \quad \times(x-1)(x+1) \quad \times(x-1)(x+1) \\ 1 &\equiv A(x+1) + B(x-1) \end{aligned}$$

- When $x = 1$,

$$\begin{aligned} 1 &= A(1+1) + B(1-1) \\ \therefore A &= \frac{1}{2} \end{aligned}$$

- When $x = -1$,

$$\begin{aligned} 1 &= A(-1+1) + B(-1-1) \\ \therefore B &= -\frac{1}{2} \end{aligned}$$

$$\therefore \frac{1}{x^2-1} = \frac{\frac{1}{2}}{x-1} - \frac{\frac{1}{2}}{x+1}$$

- ii. (3 marks)

✓ [1] for applying technique to split integrand.
 ✓ [1] for applying result from previous part.
 ✓ [1] for all correct primitives, subtracting 1 mark for each new error introduced.

$$\begin{aligned} &\int \frac{x^2+1}{x^2-1} dx \\ &= \int \frac{x^2-1+2}{x^2-1} dx \\ &= \int \left(1 + \frac{2}{x^2-1} \right) dx \\ &= \int \left(1 + 2 \left(\frac{\frac{1}{2}}{x-1} - \frac{\frac{1}{2}}{x+1} \right) \right) dx \\ &= \int \left(1 + \frac{1}{x-1} - \frac{1}{x+1} \right) dx \\ &= x + \ln(x-1) - \ln(x+1) + C \\ &= x + \ln \left(\frac{x-1}{x+1} \right) + C \end{aligned}$$

- (b) (4 marks)

✓ [1] for correct $\frac{dx}{d\theta}$ transformation.
 ✓ [1] For correct limits.
 ✓ [1] For significant progress in reducing integrand to $\sin^2 \theta \cos \theta$.
 ✓ [1] for final result.

Make the substitution $x = \tan^2 \theta$:

$$\begin{aligned} x &= \tan^2 \theta \\ \frac{dx}{d\theta} &= 2 \tan \theta \sec^2 \theta \end{aligned}$$

Altering the limits,

$$\begin{aligned} x = 0 &\quad \tan^2 \theta = 0 \\ &\quad \therefore \theta = 0 \\ x = 3 &\quad \tan^2 \theta = 3 \\ &\quad \tan \theta = \sqrt{3} \\ &\quad \theta = \frac{\pi}{3} \end{aligned}$$

Replacing x with θ :

$$\begin{aligned}
 & \int_0^3 \frac{\sqrt{x}}{(1+x)^{\frac{5}{2}}} dx \\
 &= \int_{\theta=0}^{\theta=\frac{\pi}{3}} \frac{\tan \theta}{(1+\tan^2 \theta)^{\frac{5}{2}}} \times 2 \tan \theta \sec^2 \theta d\theta \\
 &= \int_0^{\frac{\pi}{3}} \frac{2 \tan^2 \theta \sec^2 \theta}{(\sec^2 \theta)^{\frac{5}{2}}} d\theta \\
 &= \int_0^{\frac{\pi}{3}} \frac{2 \tan^2 \theta \sec^2 \theta}{\sec^3 \theta} d\theta \\
 &= 2 \int_0^{\frac{\pi}{3}} \frac{\sin^2 \theta}{\cos^2 \theta} \times \cos^3 \theta d\theta \\
 &= 2 \int_0^{\frac{\pi}{3}} \sin^2 \theta \cos \theta d\theta \\
 &= \frac{2}{3} [\sin^3 \theta]_0^{\frac{\pi}{3}} \\
 &= \frac{2}{3} \left(\frac{\sqrt{3}}{2} \right)^3 = \frac{2}{3} \times \frac{3\sqrt{3}}{8} \\
 &= \frac{\sqrt{3}}{4}
 \end{aligned}$$

(c) (4 marks)

- ✓ [1] for correct differential and alteration of limits.
- ✓ [1] for correct manipulation to quadratic trinomial in the denominator.
- ✓ [1] for applying partial fractions.
- ✓ [1] for final answer.

$$\int_0^{\frac{\pi}{2}} \frac{1}{2 - \cos x + 2 \sin x} dx$$

Making the substitution $t = \tan \frac{x}{2}$ and calculating the differential:

$$\begin{aligned}
 t &= \tan \frac{x}{2} \\
 \frac{x}{2} &= \tan^{-1} t \\
 x &= 2 \tan^{-1} t \\
 \frac{dx}{dt} &= \frac{2}{1+t^2}
 \end{aligned}$$

Altering the limits,

$$\begin{aligned}
 x = 0 & \quad t = \tan 0 \\
 x = \frac{\pi}{2} & \quad t = \tan \frac{\frac{\pi}{2}}{2} \\
 &= \tan \frac{\pi}{4} \\
 &= 1
 \end{aligned}$$

Replacing x with t :

$$\begin{aligned}
 & \int_0^{\frac{\pi}{2}} \frac{1}{2 - \cos x + 2 \sin x} dx \\
 &= \int_{t=0}^{t=1} \frac{1}{2 - \left(\frac{1-t^2}{1+t^2} \right) + 2 \left(\frac{2t}{1+t^2} \right)} \times \frac{2}{1+t^2} dt \\
 &= \int_0^1 \frac{2}{2(1+t^2) - (1-t^2) + 4t} dt \\
 &= \int_0^1 \frac{2}{3t^2 + 4t + 1} dt
 \end{aligned}$$

Decompose the integrand into partial fractions:

$$\begin{aligned}
 \frac{2}{3t^2 + 4t + 1} &= \frac{2}{(3t+1)(t+1)} \equiv \frac{A}{3t+1} + \frac{B}{t+1} \\
 \therefore 2 &\equiv A(t+1) + B(3t+1)
 \end{aligned}$$

When $t = -1$,

$$\begin{aligned}
 -2B &= 2 \\
 B &= -1
 \end{aligned}$$

When $t = -\frac{1}{3}$,

$$\begin{aligned}
 2 &= \frac{2}{3}A \\
 A &= 3
 \end{aligned}$$

Hence,

$$\begin{aligned}
 & \int_0^1 \frac{2}{3t^2 + 4t + 1} dt \\
 &= \int_0^1 \left(\frac{3}{3t+1} - \frac{1}{t+1} \right) dt \\
 &= [\ln(3t+1) - \ln(t+1)]_0^1 \\
 &= \left[\ln \left(\frac{3t+1}{t+1} \right) \right]_0^1 \\
 &= \ln \frac{4}{2} - \ln 1 = \ln 2
 \end{aligned}$$

(d) i. (3 marks)

- ✓ [1] for applying integration by parts and correctly finding the values in the u - v matrix.
- ✓ [1] for correct evaluation of $[uv]_0^1$.
- ✓ [1] for correct evaluation of $\int_0^1 v du$.

$$I_n = \int_0^1 \underbrace{x^n}_{=u} \underbrace{2^x}_{=dv} dx$$

$$u = x^n \quad v = \frac{1}{\ln 2} 2^x$$

$$du = nx^{n-1} \quad dv = 2^x$$

$$\begin{aligned} \therefore I_n &= \left[\frac{x^n 2^x}{\ln 2} \right]_0^1 - \int_0^1 nx^{n-1} \times \frac{1}{\ln 2} 2^x dx \\ &= \left(\frac{2}{\ln 2} - 0 \right) - \frac{n}{\ln 2} \int_0^1 x^{n-1} 2^x dx \\ &= \frac{2}{\ln 2} - \frac{n}{\ln 2} I_{n-1} \end{aligned}$$

ii. (3 marks)

- ✓ [1] for each of I_1 , I_2 and I_3 .

$$I_0 = \int_0^1 x^0 2^x dx = \int_0^1 2^x$$

$$= \frac{1}{\ln 2} [2^x]_0^1$$

$$= \frac{1}{\ln 2} (2 - 1) = \frac{1}{\ln 2}$$

$$I_1 = \frac{2}{\ln 2} - \frac{1}{\ln 2} I_0$$

$$= \frac{2}{\ln 2} - \frac{1}{\ln 2} \left(\frac{1}{\ln 2} \right)$$

$$= \frac{2}{\ln 2} - \frac{1}{\ln^2 2}$$

$$I_2 = \frac{2}{\ln 2} - \frac{2}{\ln 2} I_1$$

$$= \frac{2}{\ln 2} - \frac{2}{\ln 2} \left(\frac{2}{\ln 2} - \frac{1}{\ln^2 2} \right)$$

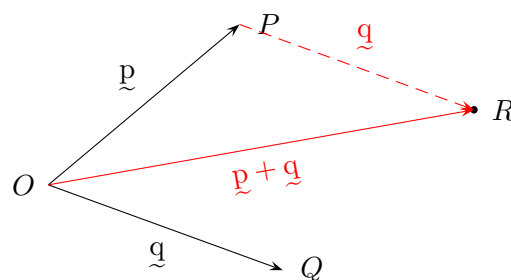
$$= \frac{2}{\ln 2} - \frac{4}{\ln^2 2} + \frac{2}{\ln^3 2}$$

$$I_3 = \frac{2}{\ln 2} - \frac{3}{\ln 2} I_2$$

$$= \frac{2}{\ln 2} - \frac{3}{\ln 2} \left(\frac{2}{\ln 2} - \frac{4}{\ln^2 2} + \frac{2}{\ln^3 2} \right)$$

$$= \frac{2}{\ln 2} - \frac{6}{\ln^2 2} + \frac{12}{\ln^3 2} - \frac{6}{\ln^4 2}$$

(a) i. (1 mark)

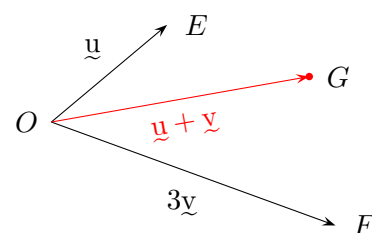


ii. (2 marks)

- ✓ [1] for reference to parallelogram and rhombus.
- ✓ [1] for correct logic and articulation of the proof.
- $OPRQ$ is a parallelogram as the addition of vectors produces a parallelogram of vectors.
- For a diagonal of a parallelogram to bisect its angles, it must be a rhombus.
- $\therefore |\underline{p}| = |\underline{q}|$ for $\overrightarrow{OR}$ to bisect $\angle POQ$.

(b) i. (2 marks)

- ✓ [1] for checking lengths of $\underline{u}$ and $\underline{v}$
- ✓ [1] for coordinates of $\overrightarrow{OG}$.



(NOT TO SCALE)

- For $\angle EOF$ to be bisected, need $|\underline{u}| = |\underline{v}|$ (result from the previous part)
- Checking lengths of $\underline{u}$ and $\underline{v}$:

$$|\underline{u}|^2 = \left| \begin{pmatrix} 2 \\ 5 \\ -7 \end{pmatrix} \right|^2 = 2^2 + 5^2 + 7^2 = 78$$

$$|\underline{v}|^2 = \left| \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \right|^2 = 5^2 + 7^2 + 2^2 = 78$$

As $|\underline{u}| = |\underline{v}|$ hence $\underline{u} + \underline{v}$ will bisect $\angle EOF$.

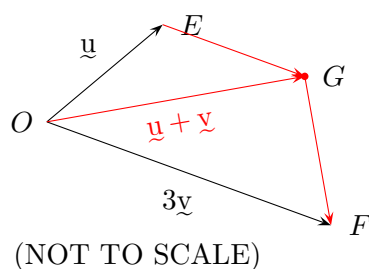
Question 3

- Hence

$$\begin{aligned}\overrightarrow{OG} &= \underline{u} + \underline{v} \\ &= \begin{pmatrix} 2 \\ 5 \\ -7 \end{pmatrix} + \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 7 \\ 12 \\ -5 \end{pmatrix}\end{aligned}$$

ii. (2 marks)

- ✓ [1] For using $\overrightarrow{EG}$ is equal to $\underline{v}$
- ✓ [1] for the shape and articulation of reasoning.

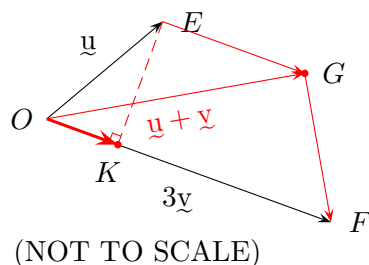


Shape is a trapezium.

- $\overrightarrow{EG} = \underline{v}$ (parallel to $\underline{v}$)
- Shape has $EG \parallel OF$ - one pair of opposite sides parallel.

iii. (3 marks)

- ✓ [1] For finding the projection of $\underline{u}$ on to $\underline{v}$.
- ✓ [1] For applying Pythagoras' Theorem to find the length of KE .
- ✓ [1] for finding the area.



Vector $\overrightarrow{OK}$ is the projection of $\underline{u}$ on

to $\underline{v}$.

$$\begin{aligned}\text{proj}_{\underline{v}} \underline{u} &= \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v} \\ &= \frac{1}{78} \left[\begin{pmatrix} 2 \\ 5 \\ -7 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \right] \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \\ &= \frac{10 + 35 - 14}{78} \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \\ &= \frac{31}{78} \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \\ \therefore |\text{proj}_{\underline{v}} \underline{u}| &= \left| \frac{31}{78} \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \right| \\ &= \left| \frac{31}{78} \right| \left| \begin{pmatrix} 5 \\ 7 \\ 2 \end{pmatrix} \right| \\ &= \frac{31}{78} \sqrt{78} = \frac{31}{\sqrt{78}}\end{aligned}$$

Applying Pythagoras' Theorem to find the perpendicular height of the trapezium KE :

$$\begin{aligned}|\text{proj}_{\underline{v}} \underline{u}|^2 + KE^2 &= |\underline{u}|^2 \\ \left(\frac{31}{\sqrt{78}} \right)^2 + KE^2 &= (\sqrt{78})^2 \\ KE^2 &= 78 - \frac{31^2}{78} = \frac{5123}{78} \\ \therefore KE &= \sqrt{\frac{5123}{78}}\end{aligned}$$

Applying the formula for the area of a trapezium,

$$\begin{aligned}A &= \frac{1}{2}h(a+b) \\ &= \frac{1}{2}(KE)(|\underline{v}| + |3\underline{v}|) \\ &= \frac{1}{2} \left(\sqrt{\frac{5123}{78}} \right) (\sqrt{78} + 3\sqrt{78}) \\ &= 2\sqrt{5123}\end{aligned}$$